Problem Set 2 — Linear Models

Statistical Modelling (BAM), IE University — Spring 2027

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Instructions. Work in pairs; each student submits their own PDF (rendered from this `.Rmd`) on Blackboard. Show derivations and interpret every number you report in context. Notation: $y = X\beta + \varepsilon$, X is $n \times p$ *including* the intercept column, $H = X(X^\top X)^{-1}X^\top$, $\hat\sigma^2 = \mathrm{RSS}/(n-p)$. Unless stated otherwise assume $\varepsilon \sim N(0, \sigma^2 I_n)$. All data sets are in R packages (`datasets`, `faraway`). Quantiles may be obtained with R (`qt`, `pt`, `qf`, `pf`). Expect 6 to 8 hours. A tidy, readable report is part of the grade: label figures, report numbers to a sensible precision, and write answers as sentences.

1 Leverage, replicated points and leave-one-out prediction

Let the linear model contain an intercept, so that $\mathbf{1} \in \mathcal{C}(X)$, and let H be its hat matrix with entries h_{ij}.

(a) Show that $H\mathbf{1} = \mathbf{1}$, so that every row of H sums to 1, and that $\sum_j h_{ij}^2 = h_{ii}$.

(b) Show that $h_{ii} = \max_{b \neq 0} (x_i^\top b)^2 / \|Xb\|^2$, where $x_i^\top$ is the i-th row of X. (Hint: $h_{ii} = \|Hd_i\|^2$ with d_i the i-th unit vector; use Cauchy–Schwarz in $\mathcal{C}(X)$.) Deduce that if m rows of X are *identical* to x_i (counting i itself), then $h_{ii} \leq 1/m$.

(c) Five observations have $x = (1, 2, 3, 4, 10)$ and $y = (2.1, 3.9, 6.2, 8.1, 16.0)$, with a simple linear regression with intercept. Compute the five leverages by hand, check that they add up to p, and flag those above $2p/n$.

(d) Compute $\hat\beta$, the residuals and $\hat\sigma^2$ by hand. Using the leave-one-out identity $y_i - x_i^\top \hat\beta_{(i)} = e_i/(1-h_{ii})$, predict y_5 from the model fitted to the other four points and give the prediction error. Check it by refitting without observation 5.

(e) Compute Cook's distance of observation 5 and explain, using $D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}$, why it is influential although its residual e_5 is the second smallest in absolute value.

2 Reading a regression table

Fertility in the 47 French-speaking Swiss provinces around 1888 (`datasets::swiss`): `Fertility` is a standardised fertility measure, `Agriculture` the percentage of males working in agriculture, `Education` the percentage of draftees with education beyond primary school, `Catholic` the percentage of Catholics and `Infant.Mortality` the percentage of live births who live less than one year. A student fits the linear model of `Fertility` on the other four variables, and a page of the printout is damaged:

```
Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       62.1013     9.6049    6.47  8.5e-08
Agriculture       -0.1546     0.0682     (A)      (B)
Education         -0.9803     0.1481     (C)  5.1e-08
Catholic             (D)      0.0289    4.31  9.5e-05
Infant.Mortality   1.0784         (E)    2.82   0.0072

Residual standard error: (F) on (G) degrees of freedom
Multiple R-squared: (H),  Adjusted R-squared: (I)
F-statistic: (J) on (K) and (L) DF,  p-value: (M)
```

You are also given: RSS $= 2158.1$, TSS $= \sum_i (y_i - \bar{y})^2 = 7178.0$; the model with `Education` and `Catholic` only has RSS $= 3054.2$; and the entries of $(X^\top X)^{-1}$ for (`Agriculture`, `Education`): $c_{AA} = 9.050 \times 10^{-5}$, $c_{EE} = 4.271 \times 10^{-4}$, $c_{AE} = 1.294 \times 10^{-4}$. Use $t_{42,0.975} = 2.018$ and R for p-values.

(a) Fill in the missing entries (A) to (M).

(b) Give a 95% confidence interval for the `Education` coefficient and interpret it in words (units and "holding fixed").

(c) Test $H_0 : \beta_{\texttt{Education}} = -1$ against the two-sided alternative.

(d) Test $H_0 : \beta_{\texttt{Agriculture}} = \beta_{\texttt{Education}}$. Show that you need the covariance of the two estimators and compute it from the information given.

(e) Test whether `Agriculture` and `Infant.Mortality` can *both* be dropped. Why is this not the same as reading the two p-values above?

(f) A newspaper writes: "Each additional percentage point of educated draftees lowers a Swiss woman's fertility by one point." Explain what is wrong, with two distinct reasons.

3 Correlated errors: what the standard error really is

Let $y = X\beta + \varepsilon$ with $\mathrm{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2 V$, where V is a known positive-definite matrix (not necessarily I).

(a) Show that $\hat{\beta}_{\mathrm{OLS}}$ is unbiased but $\mathrm{Var}(\hat{\beta}_{\mathrm{OLS}}) = \sigma^2 (X^\top X)^{-1} X^\top V X (X^\top X)^{-1}$. Show that the generalised least-squares estimator $\hat{\beta}_{\mathrm{GLS}} = (X^\top V^{-1} X)^{-1} X^\top V^{-1} y$ is obtained by ordinary least squares on the transformed data $V^{-1/2} y$, $V^{-1/2} X$, and deduce that it is BLUE with variance $\sigma^2 (X^\top V^{-1} X)^{-1}$.

(b) Now take $X = \mathbf{1}$ (intercept only, $p = 1$) and *equicorrelated* errors, $V = (1-\rho) I + \rho \mathbf{1}\mathbf{1}^\top$ with $0 < \rho < 1$ (all pairs of observations have correlation ρ, e.g. n measurements of the same unit). Show that $V\mathbf{1} = \{1 + (n-1)\rho\}\mathbf{1}$, deduce that $\hat{\beta}_{\mathrm{GLS}} = \hat{\beta}_{\mathrm{OLS}} = \bar{y}$, and that $\mathrm{Var}(\bar{y}) = \sigma^2 \{1 + (n-1)\rho\}/n$. Define the *effective sample size* $n_{\mathrm{eff}} = n / \{1 + (n-1)\rho\}$ and compute it for $n = 25$, $\rho = 0.2$.

(c) Show that $\mathrm{E}[\mathrm{RSS}] = \sigma^2 (n-1)(1-\rho)$ in this model (use $\mathrm{E}[\varepsilon^\top A \varepsilon] = \sigma^2 \operatorname{tr}(AV)$). Hence the usual estimate s^2/n of $\mathrm{Var}(\bar{y})$ has expectation $\sigma^2 (1-\rho)/n$. By what factor does it understate the true variance for $n = 25$, $\rho = 0.2$?

(d) *Simulation.* With $\sigma^2 = 1$, $n = 25$, $\rho = 0.2$ and true mean 5, generate $\varepsilon_i = \sqrt{\rho}\, z_0 + \sqrt{1-\rho}\, z_i$ with $z_0, z_1, \ldots, z_n$ iid $N(0,1)$ (new z_0 in each replication). Over 20 000 replications estimate (i) $\mathrm{Var}(\bar{y})$, (ii) $\mathrm{E}[s^2]$, (iii) the actual coverage of the nominal 95% t confidence interval `t.test(y)$conf.int`. Compare with parts (b) and (c).

(e) In two or three sentences, explain why this matters for repeated measures or clustered data, and which tool of this course handles it.

4 Collinearity in the Longley data

The data `datasets::longley` give, for 16 years (1947–1962) in the United States, `Employed` (thousands of people employed) and six economic predictors: `GNP.deflator`, `GNP`, `Unemployed`, `Armed.Forces`, `Population` and `Year`. Fit the model of `Employed` on all six predictors.

(a) Report R^2, the overall F test and the six t tests. What looks paradoxical, and why?

(b) Compute the variance inflation factor of `GNP` *by hand* (regress it on the other five predictors) and check numerically that $\widehat{\mathrm{Var}}(\hat{\beta}_{\texttt{GNP}}) = \hat{\sigma}^2 \, \mathrm{VIF} / \{(n-1) s^2_{\texttt{GNP}}\}$ matches `vcov(fit)`. Report all six VIFs.

(c) `GNP.deflator`, `GNP` and `Population` have VIFs far above 10 and large p-values. Test jointly whether the three can be removed (keep `Unemployed`, `Armed.Forces`, `Year`), and explain why this does *not* mean that "GNP has no effect on employment".

(d) Compare the two models by their leave-one-out prediction error $\mathrm{PRESS}/n = \frac{1}{n} \sum_i \{e_i/(1-h_{ii})\}^2$ and by RSS/n. What does this say about the cost of collinearity for prediction *inside* the data? Would you trust either model to extrapolate to 1975?

(e) Check the residuals for serial correlation (the observations are ordered in time) with the lag-1 autocorrelation and the Durbin–Watson statistic (`car::durbinWatsonTest`). Comment briefly.

(f) A report states: "In the Longley model the coefficient of GNP is negative (-0.036), so economic growth destroys jobs." Criticise the statement and rewrite it correctly in one or two sentences.

5 Confounding in SAT scores

The data `faraway::sat` give, for the 50 US states, the average total SAT score (`total`), the expenditure per pupil in thousands of dollars (`expend`), the pupil-teacher ratio (`ratio`), the average teacher salary in thousands (`salary`) and the percentage of students taking the test (`takers`).

(a) Let the true model be $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ and suppose we regress y on x_1 only. Show that the slope $\tilde{\beta}_1$ satisfies $\mathrm{E}[\tilde{\beta}_1] = \beta_1 + \beta_2 \delta$ where $\delta = \sum (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2) / \sum (x_{1i} - \bar{x}_1)^2$ (treating x_1, x_2 as fixed), and that, in terms of the least-squares coefficients $\hat{\beta}_1, \hat{\beta}_2$ of the *two-variable* regression, one has exactly $\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \delta$.

(b) Regress `total` on `expend` alone and then on `expend` and `takers`. Report both coefficients of `expend` with 95% confidence intervals and verify the identity in (a) numerically.

(c) Explain the sign reversal in words, in terms of how the states differ. Is `takers` a confounder, a mediator or a collider here, and does it matter for what you should adjust for?

(d) Now regress `total` on `expend`, `ratio`, `salary` and `takers`. The coefficient of `expend` becomes small and insignificant. Is expenditure irrelevant? Test jointly whether `expend`, `ratio` and `salary` can be dropped from this model, report the VIFs and explain.

(e) *Communicate.* In about 150 words, write for a state education board (non-statisticians) what these data do and do not say about whether more spending per pupil raises SAT scores. Avoid causal claims that the data cannot support, and mention one important limitation.

6 Factors and interactions: tooth growth

The data `datasets::ToothGrowth` give the length of the odontoblasts (`len`, cells responsible for tooth growth) in 60 guinea pigs, 10 for each combination of vitamin C `dose` (0.5, 1, 2 mg/day) and delivery method `supp` (`OJ` = orange juice, `VC` = ascorbic acid). Create `dosef <- factor(dose)`.

(a) Fit `len ~ supp * dosef`. Write the six cell means as functions of the six coefficients and interpret the coefficient `suppVC:dosef2` in words.

(b) Test the interaction with an F test. Then, for each dose separately, estimate the difference between VC and OJ with a 95% confidence interval (you need the variances of sums of coefficients; use `vcov()`). Summarise.

(c) Refit with `dose` as a *numeric* predictor, `len ~ supp * dose`, and compare it with the factor model by a nested F test. What do you conclude about the dose–response relationship? Try `log2(dose)` and comment.

(d) The design is balanced. Show with `anova()` that the sequential sums of squares of `supp` and `dosef` in `len ~ supp + dosef` do not depend on the order of the terms, and explain why this is *not* the case for unbalanced data such as `mtcars` with `mpg ~ wt + hp`.

(e) A colleague proposes `len ~ dose + supp:dose` (no main effect of `supp`) "because supp has no effect at dose 0". Which principle does this violate, and why can it be harmful here (dose has no value zero in the data)?

7 Model selection and prediction: body fat

The data `faraway::fat` contain body measurements of 252 men. Take the response `brozek` (percentage of body fat) and the 13 predictors `age`, `weight`, `height`, `neck`, `chest`, `abdom`, `hip`, `thigh`, `knee`, `ankle`, `biceps`, `forearm`, `wrist` (do not use `siri`, `density`, `adipos` or `free`, which are computed from the response or from body weight and are not available when predicting).

(a) *Theory.* Suppose σ^2 is known and the errors are normal. Show that ranking submodels by Mallows' $C_p = \mathrm{RSS}_p/\sigma^2 - n + 2p$ is the same as ranking them by AIC. What changes when σ^2 is replaced by $\hat{\sigma}^2_{\text{full}}$?

(b) Use `leaps::regsubsets` to find the best subset of each size. Which size minimises C_p, and which minimises BIC? Report the variables chosen by BIC and explain why the two criteria disagree.

(c) Run `step()` with AIC and with BIC (penalty $\log n$) from the full model and report the selected models. Report the 95% confidence interval of the `abdom` coefficient in the full model and in the BIC-selected model. Why is the second interval *not* a valid 95% interval?

(d) *Honest comparison.* Set a seed, hold out 80 men at random as a test set, and on the remaining 172 fit (i) the full model, (ii) the step-AIC model, (iii) the step-BIC model, (iv) a lasso with `cv.glmnet` (fixed `foldid`, `lambda.min`). Selection, standardisation and the choice of λ must use the training rows only. Compute the test MSE of each. Repeat for 100 random splits and report mean and standard deviation of each test MSE. What do you conclude?

(e) Run the standard diagnostics for the full model on all 252 men. Identify the cases with the largest leverage and the largest Cook's distance, and look at their rows. Is there evidence of data errors?

(f) Two clients: a gym that wants to *predict* body fat from tape measurements, and a physician who wants to estimate how much body fat increases per cm of abdominal circumference *adjusting for age and weight*. What model-building strategy do you recommend for each?

8 Ridge regression and the lasso

Let X be $n \times p$ with centred columns and y centred; ridge regression minimises $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$.

(a) Show that $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ is the ordinary least-squares estimate for the *augmented* data $\tilde{y} = (y^\top, 0_p^\top)^\top$, $\tilde{X} = (X^\top, \sqrt{\lambda} I_p)^\top$, and that it exists for every $\lambda > 0$ even when $\operatorname{rank} X < p$. With the SVD $X = UDV^\top$ show that the fitted values are $\hat{y}_\lambda = \sum_j u_j \frac{d_j^2}{d_j^2 + \lambda} (u_j^\top y)$ and that the effective degrees of freedom $\operatorname{tr}(S_\lambda)$, $S_\lambda = X(X^\top X + \lambda I)^{-1} X^\top$, equal $\sum_j d_j^2/(d_j^2 + \lambda)$.

(b) *Leave-one-out shortcut.* Write the ridge fit as $\hat{y} = Sy$ ($S = S_\lambda$ for centred data; $S = S_\lambda + \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ if an unpenalised intercept is estimated), let $\hat{y}_i^{(-i)}$ be the ridge prediction of y_i from the fit without observation i, and s_{ii} the diagonal of S. Show that $y_i - \hat{y}_i^{(-i)} = (y_i - \hat{y}_i)/(1 - s_{ii})$. (Hint: show that refitting with y_i replaced by $\hat{y}_i^{(-i)}$ gives the same coefficients as the fit without i.)

(c) *Lasso, orthonormal case.* Suppose $X^\top X/n = I_p$ and the least-squares estimates are $z = (3.0, -1.5, 0.8, -0.2, 0.05)$. Derive the minimiser of $\frac{1}{2n}\|y - X\beta\|^2 + \lambda\|\beta\|_1$ and compute the lasso and ridge (the glmnet ridge, $\hat{\beta}/(1 + \lambda)$) estimates for $\lambda = 0.6$. For which λ does the lasso select exactly three variables?

(d) *In R (Longley).* Standardise the six predictors of `longley` (`scale`) and centre `Employed`. (i) Compute the ridge estimate for $\lambda = 1$ by the closed form and by least squares on the augmented data. (ii) Compute the singular values of X and the effective degrees of freedom for $\lambda = 0.01, 0.1, 1, 10$. (iii) Compute the leave-one-out error of (b) on a grid of λ from 10^{-4} to 10 (the smoother of the model with an unpenalised intercept is $S_\lambda + \frac{1}{n}\mathbf{1}\mathbf{1}^\top$) and verify the shortcut by brute force for one λ. (iv) Choose λ by the shortcut, report the coefficients against OLS and comment.

(e) Explain why the predictors must be standardised before applying the penalty, and what would happen to the ridge estimate of a coefficient if its predictor were measured in metres instead of centimetres.